Tap here for Join Aakash ans key

30/09/2026

Code-D Phase-4

Aakash
Medical | IIT-JEE | Foundations

Corporate Office : AESL, 3rd Floor, Incuspaze Campus-2, Plot No. 13, Sector-18, Udyog Vihar, Gurugram, Haryana - 122015, ***Ph.***+91-1244168300

MM : 720

Fortnightly Test for NEET-2027 (XII-Passed_P4)_FT-02D

Time : 180 Min.

Tap here for Join Aakash ans key

PHYSICS

1. (4)	**2.** (1)	**3.** (4)	(2)	**5.** (1)	**6.** (2)
7. (2)	**8.** (1)	**9.** (3)	(2)	**11.** (4)	**12.** (1)
13. (1)	**14.** (1)	**15.** (1)	(3)	**17.** (1)	**18.** (3)
19. (4)	**20.** (1)	**21.** (1)	(2)	**23.** (3)	**24.** (3)
25. (2)	**26.** (2)	**27.** (1)	(4)	**29.** (2)	**30.** (2)
31. (2)	**32.** (3)	**33.** (3)	(2)	**35.** (1)	**36.** (4)
37. (4)	**38.** (4)	**39.** (2)	(4)	**41.** (4)	**42.** (2)
43. (2)	**44.** (3)	**45.** (1)	**Your paragraph text**		

CHEMISTRY

46. (4)	**47.** (3)	**48.** (3)	**49.** (3)	**50.** (2)	**51.** (4)
52. (1)	**53.** (1)	**54.** (2)	**55.** (1)	**56.** (1)	**57.** (4)
58. (1)	**59.** (2)	**60.** (1)	**61.** (1)	**62.** (2)	**63.** (1)
64. (3)	**65.** (4)	**66.** (4)	**67.** (1)	**68.** (4)	**69.** (4)
70. (2)	**71.** (4)	**72.** (1)	**73.** (1)	**74.** (4)	**75.** (2)
76. (1)	**77.** (1)	**78.** (2)	**79.** (1)	**80.** (4)	**81.** (3)
82. (1)	**83.** (4)	**84.** (4)	**85.** (1)	**86.** (1)	**87.** (4)
88. (1)	**89.** (1)	**90.** (4)			

Tap here for Join Aakash ans key

BOTANY

91. (3)	**92.** (2)	**93.** (2)	**94.** (3)	**95.** (3)	**96.** (2)
97. (2)	**98.** (3)	**99.** (2)	**100.** (4)	**101.** (2)	**102.** (3)
103. (1)	**104.** (2)	**105.** (1)	**106.** (2)	**107.** (3)	**108.** (3)
109. (1)	**110.** (2)	**111.** (4)	**112.** (4)	**113.** (1)	**114.** (1)
115. (1)	**116.** (4)	**117.** (3)	**118.** (1)	**119.** (4)	**120.** (4)
121. (1)	**122.** (2)	**123.** (4)	**124.** (1)	**125.** (3)	**126.** (1)
127. (3)	**128.** (3)	**129.** (1)	**130.** (4)	**131.** (3)	**132.** (2)
133. (3)	**134.** (3)	**135.** (1)			

ZOOLOGY

136. (1)	**137.** (1)	**138.** (2)	**139.** (1)	**140.** (2)	**141.** (1)
142. (2)	**143.** (3)	**144.** (4)	**145.** (1)	**146.** (2)	**147.** (4)
148. (4)	**149.** (3)	**150.** (4)	**151.** (2)	**152.** (1)	**153.** (1)
154. (2)	**155.** (3)	**156.** (2)	**157.** (1)	**158.** (4)	**159.** (3)
160. (4)	**161.** (4)	**162.** (3)	**163.** (4)	**164.** (1)	**165.** (3)
166. (4)	**167.** (1)	**168.** (4)	**169.** (1)	**170.** (3)	**171.** (3)
172. (3)	**173.** (3)	**174.** (4)	**175.** (1)	**176.** (3)	**177.** (1)
178. (2)	**179.** (3)	**180.** (2)			

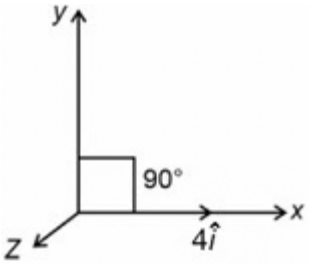
Hints and Solutions

PHYSICS

(1) **Answer :** (4)
Solution:
A vector is not changed if it slides parallel to itself.

(2) **Answer :** (1)
Solution:
Only statement I is correct while statement II is incorrect because $|\vec{\lambda A}| \neq \lambda |\vec{A}|$ if λ is a negative number.

(3) **Answer :** (4)
Solution:
$\vec{A} + \vec{B} = 4i + 2j - 2j + 5k - 5k = 4i$



(4) **Answer :** (2)
Hint:
Use equation of motion, $\vec{v} = \vec{u} + a\vec{t}$
Solution:
$\vec{u} = (i - 2j)\ \text{m s}^{-1}$ and $\vec{a} = (-i + 2j)\ \text{m s}^{-2}$

$\vec{v} = \vec{u} + a\vec{t}$
$\vec{v} = (i - 2j) + (-i + 2j) \times 2$
$\vec{v} = (i - 2j) + (-2i + 4j)$
$\vec{v} = (-i + 2j)\ \text{m s}^{-1}$

(5) **Answer :** (1)
Solution:
$\vec{v}_{S/R} = \vec{v}_S - \vec{v}_R$
$\vec{v}_S = \vec{v}_{S/R} + \vec{v}_R$

In case of minimum time, angle between $\vec{v}_{S/R}$ and $\vec{v}_R$ is 90°.
$v_s = \sqrt{8^2 + 4^2}$
$= \sqrt{80}$
$\simeq 9$ m/s

(6) **Answer :** (2)
Solution:
$20\sqrt{2} \cos 60° = v \cos 45°$
$20\sqrt{2} \times \frac{1}{2} = v \times \frac{1}{\sqrt{2}}$
$v = 20\ \text{m s}^{-1}$

(7) **Answer :** (2)
Solution:

Aakash Medical|IIT-JEE|Foundations

$|\vec{V}_{el}| = \sqrt{v^2 + v^2 - 2v^2\cos 90°}$

$= v\sqrt{2}$

(8) **Answer :** (1)
Solution:

$y = ax - \frac{bx^2}{2}$

$y = x\tan\theta\left[1 - \frac{x}{R}\right]$

$y = ax\left[1 - \frac{bx}{2a}\right]$

So, $R = \boxed{\frac{2a}{b}}$

(9) **Answer :** (3)
Solution:

$\vec{u}\,A = i + j \quad \vec{u}\,B = i \quad \vec{u}\,C = i - j$

$uA = \sqrt{2}, \quad uB = 1, \quad uC = \sqrt{2}$

$\frac{u_B}{u_C} = \frac{1}{\sqrt{2}}$

(10) **Answer :** (2)
Solution:

$\frac{R_{max}}{4} = H_{max}$ (at $\theta = 45°$)

$\therefore$ Coordinates of max height (100, 50)

(11) **Answer :** (4)
Solution:
Circular motion is not possible without centripetal acceleration. For non-uniform circular motion both *ar* and *at* are present.

(12) **Answer :** (1)
Solution:

$\omega = 2\pi n$

$= 2\pi \times 140$

$= 2 \times \frac{22}{7} \times 140 = 880$ rad /s

(13) **Answer :** (1)
Solution:

$a = \omega^2 r$

$\frac{a_2}{a_1} = \left(\frac{\omega_1}{\omega_2}\right)^2 = \left(\frac{2}{4}\right)^2 = \frac{1}{4}$

(14) **Answer :** (1)
Solution:
Speed at maximum height $= u\cos\theta = \frac{u}{2}$

$\Rightarrow \cos\theta = \frac{1}{2} \Rightarrow \theta = 60°$

(15) **Answer :** (1)
Solution:

$a = \frac{F}{m} = \frac{16}{8} = 2$ m/s^2

(16) **Answer :** (3)
Solution:
External unbalanced force will accelerate the body.

(17) **Answer :** (1)
Solution:

$\vec{F} = 3i + 4j - 5k$

$F = \sqrt{3^2 + 4^2 + (-5)^2} = 5\sqrt{2}$ N

$F = Ma$

$5\sqrt{2} = M\sqrt{2}$

$M = 5$ kg

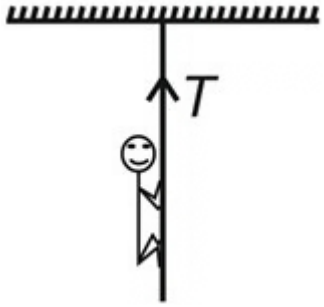
Aakash Medical|IIT-JEE|Foundations

(18) **Answer :** (3)
Solution:
Impulse $= |\vec{\Delta p}| = MV - (-MV) = 2MV$

(19) **Answer :** (4)
Solution:
$\vec{F} = \frac{d\vec{P}}{\vec{F}}$
If $_{ext} = 0$, then $\Delta\vec{P} = 0$
Internal forces always cancel each other and hence cannot change the momentum of the body/system.

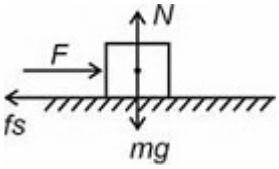
(20) **Answer :** (1)
Solution:
$v = \sqrt{\mu Rg} = \sqrt{0.5 \times 13 \times 10}$
$= \sqrt{65} = 8.06 \approx 8\ m/s$

(21) **Answer :** (1)
Hint:
$F = Ma$
Solution:



$T - mg = ma$
$T = mg + ma$
$= 40(9.8 + 2)$
$= 40 \times 11.8$
$= 472$ N

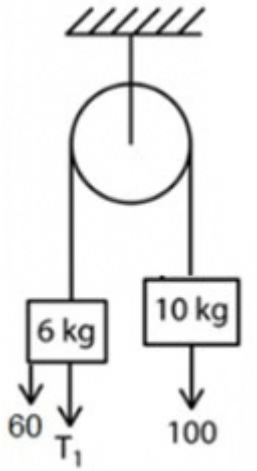
(22) **Answer :** (2)
Hint:
Block will move only if $F > (fs)_{max}$
Solution:



Here, $(fs)_{max} = \mu s \times N$
$= 0.4 \times 50$
$= 20$ N
$F < (fs)_{max}$, so block will not move.
Hence, friction on block, $f = F = 16$ N

(23) **Answer :** (3)
Solution:

Aakash Medical|IIT-JEE|Foundations



For equilibrium of 6 kg block

100 = 60 + $T1$

$T1$ = 40 N

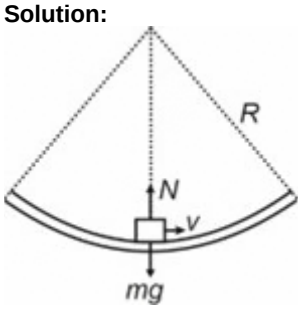
(24) **Answer :** (3)

Solution:

$v = \frac{0.1}{2} \times (200) = 10 \text{m/s}$

(25) **Answer :** (2)

Solution:



$N - mg = \frac{mv^2}{R}$

$N = mg + \frac{mv^2}{R}$

Aakash Medical|IIT-JEE|Foundations

(26) **Answer :** (2)

Solution:

$F = \frac{vdm}{dt}$

F = 4 ×2=8 N

(27) **Answer :** (1)

Solution:

$\vec{F}net = \frac{dp_{system}}{dt}$

$\vec{F}net = 0$

$\frac{d\vec{P}_{system}}{dt} = 0$

$\vec{P}system$ = constant or $\vec{P}_{system} = 0$ and $\vec{a}_{com} = 0$

(28) **Answer :** (4)

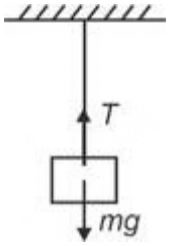
Solution:

$F = \frac{dp}{dt} \Rightarrow \int dp = \int F dt$

$I = \Delta P = \int F dt$

(29) **Answer :** (2)

Solution:



Here tension is acting away from the object, therefore it is pulling force.

$\sum \vec{F} = 0$

$T - mg = 0$

$T = mg$

Static friction is a self adjusting force.

(30) **Answer :** (2)

Solution:

$\frac{P^2}{2m}$

$-\frac{1}{2}mv^2 = 4mv$

$v = 8$ unit

(31) **Answer :** (2)

Solution:

$F = 5x$

$W = \int Fdx = \int_{x=0}^{x=2} 5xdx$

$W = 5\left[\frac{x^2}{2}\right]_0^2 = \frac{5}{2}[2^2 - 0^2] = 10$ J

(32) **Answer :** (3)

Solution:

$h = e^2 h_o$

$= (0.6)2\ (100) = 36$ m

(33) **Answer :** (3)

Solution:

$\text{Power} = \frac{\text{Work}}{\text{Time}} \Rightarrow 5\times10^3 = \frac{mgh}{3600}$

$m \Rightarrow \frac{5\times10^3\times3600}{50\times10} = 36000$kg

(34) **Answer :** (2)

Solution:

$\Delta U = \Delta K$

$= \frac{1}{2}\times0.1\times20^2 = 20$ J

(35) **Answer :** (1)

Solution:

$U = \frac{1}{2}Kx^2 = \frac{1}{2}\times K\times\left(\frac{F}{K}\right)^2 = \frac{F^2}{2K}$

For constant force, $U1K1 = U2K2$

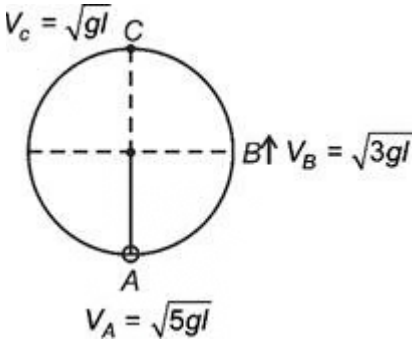
$\frac{U_1}{U2} = \frac{K_2}{K1} = \frac{200}{100} = 2$

(36) **Answer :** (4)

Solution:

In critical case of vertical circular motion,
we have

Aakash Medical|IIT-JEE|Foundations



(37) **Answer :** (4)
Solution:
Friction is a non-conservative force.

(38) **Answer :** (4)
Hint:
Use, $F = -\frac{dU}{}$

Solution:
$F = -2x + 5$
$F(x = 1\text{ m}) = -2 + 5 = 3\text{ N}$

(39) **Answer :** (2)
Solution:
W.D. by conservative force = $-\Delta U$
ΔU = –ve

(40) **Answer :** (4)
Solution:
$t \propto u$
$\frac{t_1}{t_2} = \frac{u_1}{u}$
$\frac{2}{t_2} = \frac{36}{144} \Rightarrow t2 = 8\text{ s}$

(41) **Answer :** (4)
Hint:
Use conservation of linear momentum
Solution:
$\vec{P}_i = \vec{P}_f$
$\Rightarrow 6 - 4 = 4vc$
$\Rightarrow v = \frac{1}{2}\text{ m s}^{-1}$

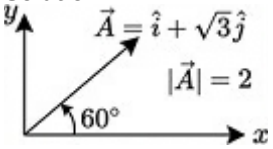
(42) **Answer :** (2)
Solution:

e = coefficient of restitution = $\frac{\text{VOS}}{\text{VOA}} = \frac{\text{Vel. of separation}}{\text{Vel. of approach}}$

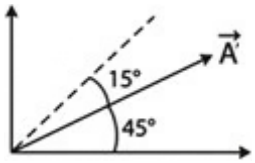
In case inelastic collision.
Colliding bodies stick together.
$\therefore e = 0$

(43) **Answer :** (2)
Solution:
Work done by all the forces is equal to change in kinetic energy.

(44) **Answer :** (3)
Solution:



Aakash Medical|IIT-JEE|Foundations



$A_x = 2\cos 45^\circ$

$A_y = 2\sin 45^\circ$

New vector $\vec{A'} = \sqrt{2}i + \sqrt{2}j = \sqrt{2}i + \sqrt{2}j$.

$\frac{\vec{A'}}{|\vec{A'}|} = A = \frac{\sqrt{2}i + \sqrt{2}j}{} = \frac{1}{\sqrt{2}}i + \frac{1}{\sqrt{2}}j.$

(45) **Answer :** (1)

Solution:

$P = 2t + 3t2$

$\frac{dP}{dt} = 2 + 6t$

$F = 2 + 6 \times 2 = 14$ N

CHEMISTRY

(46) **Answer :** (4)

Solution:

The oxidation state of Al is +3 and covalency is 6.

$x - 1 + 0 \times 5 = +2$

$x = +3$

(47) **Answer :** (3)

Solution:

p-Block elements along with *s*-block elements are known as representative elements.

(48) **Answer :** (3)

Solution:

Given electronic configuration is of Ga(31).

The atomic number of the element just above it is 31 – 18 = 13

(49) **Answer :** (3)

Solution:

Be is in diagonal relationship with Al

(50) **Answer :** (2)

Solution:

Z = 105 : Unnilpentium (Unp).

(51) **Answer :** (4)

Solution:

Element with atomic number 33 is **Arsenic**.

[Ar] 3*d*10 4*s*2 4*p*3

period number = maximum value of principal quantum number = 4

Group number = 10 + 5 = 15

(52) **Answer :** (1)

Solution:

Chlorine has highest negative electrons gain enthalpy which belong to 3rd period group 17 in periodic table.

(53) **Answer :** (1)

Solution:

$O(g) + e^- \rightarrow O^-$, $\Delta H^\circ = -142$ kJ mol–1

But O^- resists the addition of another electron and $O^-(g) + e^- \rightarrow O2^-(g)$, $\Delta H^\circ = 844$ kJ mol–1

(54) **Answer :** (2)

Hint:

The normal oxide formed by the element on extreme left is most basic, whereas that formed by the element on extreme right of periodic table is the most acidic.

Solution:

Cl_2O_7 → Acidic

As_2O_3 → Amphoteric

N_2O → Neutral

Na_2O → Basic

(55)

Answer : (1)

Solution:

Ununoctium (118) → Oganesson
Ununhexium (116) → Livermorium
Unnilennium (109) → Meitnerium
Unniltrium (103) → Lawrencium

(56)

Answer : (1)

Solution:

$Na \rightarrow Na^+ + e^-$ E = 5.1 eV

$Na^+ + e^- \rightarrow Na$ E = –5.1 eV

(57) **Answer :** (4)

Solution:

For isoelectronic species, higher is the negative charge, larger is the size of ion

(58) **Answer :** (1)

Solution:

Arsenic is a metalloid among group 15 elements.

(59) **Answer :** (2)

Solution:

There are 7 horizontal rows called as periods and 18 vertical columns called as groups in the modern periodic table.

(60) **Answer :** (1)

Solution:

The ionisation energy order of 3rd period elements given here is

Si
< P
> S
<
Cl

The value is high P over S is due to extra stable *ns*2*np*3 configuration in P.

(61) **Answer :** (1)

Solution:

Element	Atomic radii/pm
Li	152
Be	111
Na	186
Mg	160

(62) **Answer :** (2)

Solution:

According to Dobereiner's traid, the middle element of each of the triads had an atomic weights about half way between the atomic weights of the other two elements. Dobereiner's traid has similar physical and chemical properties.

(63) **Answer :** (1)

Solution:

Eka-aluminium was predicted by Mendeleev which is now called as gallium.

(64) **Answer :** (3)

Hint:

$\frac{1}{\lambda} = R_H \times Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$

Solution:

For series limit of Paschen series, n1 = 3 and n2 = ∞

$\frac{1}{\lambda} = R_H \times Z^2 \left(\frac{1}{32} - \frac{1}{\infty 2}\right) = \frac{R_H}{9}$

$\Rightarrow \lambda = \frac{9}{R_H}$

(65) **Answer :** (4)
Solution:

$\lambda = \frac{h}{\sqrt{2mK}}$ where h and m (mass) are constants

$\frac{\lambda_1}{\lambda 2} = \sqrt{\frac{K_2}{K_1}}$

$K_2 = 3K_1$

$\frac{\lambda_1}{\lambda 2} = \sqrt{\frac{3K_1}{K_1}}$

$\frac{\lambda 1}{\lambda_2} = \sqrt{3}$

$\lambda 2 = \frac{\lambda 1}{\sqrt{3}}$

(66) **Answer :** (4)
Solution:
Alpha particle is not a subatomic particle.

(67) **Answer :** (1)
Solution:
For multielectronic system, greater is the sum of (n + l), greater is the energy of orbitals.
In case, two subshell have same (n + l) values then the subshell with greater n value will have more energy
Value of (n + l) :
4*f* = 4 + 3 = 7
5*p* = 5 + 1 = 6
4d = 4 + 2 = 6
4*p* = 4 + 1 = 5
∴ The order of energy of orbital is
4*f* > 5*p* > 4*d* > 4*p*

(68) **Answer :** (4)
Solution:
For a given l value, m can be
–l, –(l – 1) ……0, 1 ….. (l – 1), l
–l to +l including zero.

(69) **Answer :** (4)
Solution:
n = 4, l = 3 is 4f subshell, has 14 electrons at most.

(70) **Answer :** (2)
Solution:

Orbital	Number of radial nodes (n – l–1)
2*p*	2 – 1 – 1 = 0
3*p*	3 – 1 – 1 = 1
3*s*	3 – 0 – 1 = 2
4*s*	4 – 0 – 1 = 3

(71) **Answer :** (4)
Hint:
Chromium and copper have exceptional electronic configuration.
Solution:

29Cu = 1*s*2 2*s*2 2*p*6 3*s*2 3*p*6 3*d*10 4*s*1

24Cr = 1*s*2 2*s*2 2*p*6 3*s*2 3*p*6 3*d*5 4*s*1

25Mn = 1*s*2 2*s*2 2*p*6 3*s*2 3*p*6 3*d*5 4*s*2

Aakash Medical|IIT-JEE|Foundations

26Fe = 1*s*2 2*s*2 2*p*6 3*s*2 3*p*6 3*d*6 4*s*2

Element	Unpaired electron(s)
Cu	1
Mn	5
Cr	6
Fe	4

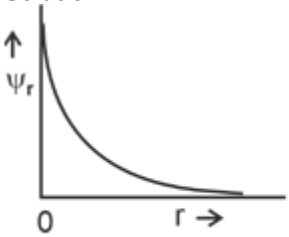
(72) **Answer :** (1)
Hint:
Orbital angular momentum $=\sqrt{l(l+1)}\,\hbar$
Solution:
Orbital angular momentum for
2*p* orbital : $\sqrt{1(1+1)}\,\hbar \Rightarrow \sqrt{2}\,\hbar$
3*d* orbital : $\sqrt{2(2+1)}\,\hbar \Rightarrow \sqrt{6}\,\hbar$
4*s* orbital : $\sqrt{0(0+1)}\,\hbar \Rightarrow 0$
4*f* orbital : $\sqrt{3(3+1)}\,\hbar = 2\sqrt{3}\,\hbar$

(73) **Answer :** (1)
Solution:
K.E of electron $=13.6\times\frac{Z^2}{n^2} \Rightarrow K.E \propto n^{-2}$
Angular momentum $=mvr=\frac{nh}{2\pi}$
i.e. Angular momentum $\propto$ n
Radius of nth orbit $=0.529\times\frac{n^2}{Z}$
i.e. $r \propto n^2$
Velocity of electron in nth orbit $=2.188\times\frac{Z}{n}\times10^6$ m/s
i.e. $v \propto \frac{1}{n} \propto n^{-1}$

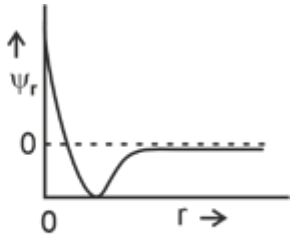
(74) **Answer :** (4)
Solution:
The correct order of their metallic character is
K > Mg > Al > B

(75) **Answer :** (2)
Solution:
Power of bulb = 30 watt = 30 Js^{-1}
Energy of photon $=\frac{hc}{\lambda}=\frac{6.6\times10^{-34}\times3\times10^8}{1320\times10^{-9}}=1.5\times10^{-19}$ J
Number of photons emitted per second $=\frac{30}{1.5\times10^{-19}}=20\times10^{19}=2\times10^{20}$

(76) **Answer :** (1)
Solution:

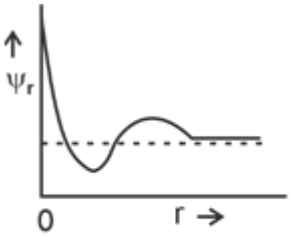


(1*s*); number of radial node = 0

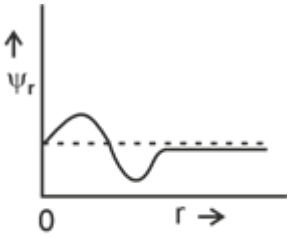


(2*s*); number of radial node = 1

Aakash Medical|IIT-JEE|Foundations



(3*s*); number of radial nodes = 2



(3*p*); number of radial node = 1

(77) **Answer :** (1)
Solution:

• Mass of electron = 9.1 × 10–31 kg

• Mass of proton = 1.6726 × 10–24 kg

• Mass of neutron = 1.6749 × 10–27 kg

• α-particle is He2+ which contains 2 proton and 2 neutron.

(78) **Answer :** (2)
Hint:
Electron is more tightly bound to the nucleus in smallest allowed orbit.
Solution:
electron has more negative energy in n = 1 than n = 5.
∴ electron is tightly bound to nucleus in n = 1.
For a subshell, maximum number of electrons are 4l+2, where l = azimuthal quantum number

(79) **Answer :** (1)
Solution:

$\Delta x \Delta p = \frac{h}{4\pi}$

$\Delta x = \frac{h}{4\pi \times \Delta p}$

$= \frac{6.626 \times 10^{-34}}{4 \times 3.14 \times 5 \times 10^{6}}$

= 0.105 × 10–28 m

= 1.05 × 10–29 m

Answer : (4)

(80) **Hint:**

Potential energy = 2 × $\left(-13.6 \frac{Z^2}{n2}\right) eV$

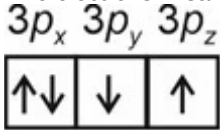
Solution:

$\frac{(PE)_{He^+}}{(PE)_{Be^{3+}}} = \frac{2 \times -13.6 \times \frac{(2)}{2}}{2 \times -13.6 \times \frac{(4)}{2}}$

$= \frac{(2)^2 \times 2}{(2)(4)^{-2} \times (4)} = \frac{1}{16}$ $\frac{(4)}{2}$ $\frac{(2)}{2}$

(81) **Answer :** (3)
Solution:

ψ does not have physical meaning but ψ2 does.
Two electrons in same orbital can be distinguished by spin quantum number only.



For the two electrons present in 3px orbital the n, ℓ and mℓ values are same
n = 3
ℓ = 1
mℓ = +1, 0, –1

(82) **Answer :** (1)

Aakash Medical|IIT-JEE|Foundations

Solution:

$\frac{1}{\lambda} = R\left[\frac{1}{22} - \frac{1}{n^2}\right] = R\left[\frac{n^2-4}{4n^2}\right]$

$\lambda = \frac{4}{R}\left[\frac{n2}{n2-4}\right]$

$\lambda = k \times \frac{n2}{n2-4}$ so $k = \frac{4}{R}$

(83) **Answer :** (4)
Solution:

$KE = h\nu - h\nu_0$

$= 6.6 \times 10{-}34(1 \times 1016 - 0.2 \times 1016)$

$= 6.6 \times 10{-}34 \times 0.8 \times 1016$

$= 5.28 \times 10^{-18}\ J \approx 5.3 \times 10^{-18}\ J$

(84) **Answer :** (4)
Solution:
The correct order of wavelength of given electromagnetic waves is radiowaves > Microwave > IR > UV

(85) **Answer :** (1)
Solution:

$\nu = \frac{c}{\lambda}$

$V = \frac{\nu c}{} = \frac{7.5 \times 10^{14}}{3 \times 10^8}$

$= 2.5 \times 106\ m{-}1$

(86) **Answer :** (1)
Solution:
Diffraction and interference are explained by wave nature of electromagnetic radiations.

(87) **Answer :** (4)
Solution:
Double ionized helium atom *i.e.*, helium nuclei was impinged on a metal foil in Rutherford's nuclear model.

(88) **Answer :** (1)
Solution:
Size of shell is given by principal quantum number.

(89) **Answer :** (1)
Solution:
Isobars are the atoms with same mass number but different atomic number.

(90) **Answer :** (4)
Solution:
• For Lyman series, n1 = 1 and n2 = 2, 3 …. Since electron jumps to first excited state (n = 2) hence no line will be observed in Lyman series.
• For Paschen series, n1 = 3 and n2 = 4, 5 …. Hence number of lines obtained will be 3.
• Total number of lines obtained will be 10.

BOTANY

(91) **Answer :** (3)
Solution:
At the end of meiosis, each daughter cell is haploid and has 4 chromosomes. Therefore, the original diploid cell before meiosis would have twice the number of chromosomes, as meiosis reduces the chromosome number by half. Thus, the diploid cell in the G1 phase of the cell cycle would have 2 × 4 = 8 chromosomes.

(92) **Answer :** (2)
Solution:
In the telophase, reappearance of nuclear membrane can be observed.

(93) **Answer :** (2)
Solution:

Dissolution of the synaptonemal complex occurs During Diplotene stage of Prophase-I of Meiosis-I.

(94) **Answer :** (3)
Hint:
During the second phase of karyokinesis, mitotic spindle formation is completed.
Solution:
The microtubules of the spindle fibres attach to the kinetochore during the metaphase.

(95) **Answer :** (3)
Solution:
A bivalent in prophase I of meiosis consists of two homologous chromosomes, each made up of two sister chromatids. Therefore, it represents two centromeres with four chromatids.

(96) **Answer :** (2)
Solution:
In animal cell, centriole duplication occurs during S-phase.

(97) **Answer :** (2)
Solution:
DNA replication and duplication of centrioles are the events associated with S phase.

(98) **Answer :** (3)
Solution:
Four haploid cells are formed at the end of meiosis II from a diploid mother cell.

(99) **Answer :** (2)
Hint:
Prophase is the longest phase of M-phase.
Solution:
The interphase lasts more than 95% of the duration of cell cycle.

(100) **Answer :** (4)
Solution:
During cell division, karyokinesis is followed by cytokinesis.

(101) **Answer :** (2)
Solution:
Splitting of centromeres and separation of chromatids occur during anaphase.

(102) **Answer :** (3)
Solution:
In the quiescent stage (G0), cells remain metabolically active but no longer proliferate.

(103) **Answer :** (1)
Hint:
Cell growth results in disturbing the ratio between the nucleus and cytoplasm.
Solution:
It is essential for the cell to divide to restore the nucleo-cytoplasmic ratio.

(104) **Answer :** (2)
Solution:
Interphase stage of cell cycle involves growth of cell and does not involves division of cell.

(105) **Answer :** (1)
Hint:
Liquid endosperm in coconut is multinucleated in nature.
Solution:
In some organisms karyokinesis is not followed by cytokinesis as a result of which multinucleated condition arises.

(106) **Answer :** (2)
Solution:
Newly formed daughter cells can themselves grow and divide, giving rise to a new cell population that is formed by the growth and division of a single parental cell and its progeny.

(107) **Answer :** (3)
Solution:
The activity of recombinase enzyme is observed during pachytene stage of prophase I.

(108) **Answer :** (3)
Solution:

Aakash Medical|IIT-JEE|Foundations

The two asters together with spindle fibre form mitotic apparatus. Each centrosome radiates out microtubule called ‘asters’.

(109)Answer : (1)

Hint:

No recombination of genetic material occur during mitosis.

Solution:

Meiosis produces gametes for sexual reproduction and also genetic variations in the generations.

(110)Answer : (2)

Solution:

Prophase is marked by the initiation of condensation of chromosomal material.

(111)Answer : (4)

Hint:

It is preparatory phase.

Solution:

Cell cycle has two phases *i.e*., interphase and mitotic phase. Interphase comes between two successive M phases, called preparatory phase.

(112)Answer : (4)

Hint:

Mitosis helps to repair the lining of gut.

Solution:

Meiocytes (PMC) are the cells that undergo meiosis to produce gametes.

(113)Answer : (1)

Solution:

The order of cats and dogs is carnivora.

(114)Answer : (1)

Solution:

According to the rules of binomial nomenclature of organisms, both genus and species are underlined if handwritten, and both are in italics if printed.

(115)Answer : (1)

Solution:

Division-Angiospermae

(116)Answer : (4)

Solution:

The correct sequence of taxonomic categories is as follows:
Kingdom → Division → Class → Order → Family → Genus → Species

(117)Answer : (3)

Solution:

Datura is placed in family Solanaceae

(118)Answer : (1)

Solution:

The lowest and basic category of classification is species.

(119)Answer : (4)

Solution:

Taxonomy does not include phylogeny or evolutionary history.

(120)Answer : (4)

Solution:

The word systematics is derived from the Latin word ‘systema’ which means systematic arrangement of organisms.

(121)Answer : (1)

Hint:

Meiosis II is similar to mitosis.

Solution:

The correct sequential order is
A (Prophase II) → D (Metaphase II) → C (Anaphase II) → B (Telophase II)

(122)Answer : (2)

Hint:

Sole members of kingdom Monera are bacteria.

Solution:

Bacteria are structurally simple but very complex in behaviour.

(123)Answer : (4)
Solution:
Halophiles are found in extreme saline conditions.

(124)Answer : (1)
Solution:
Coccus – Spherical
Bacillus – Rod-shaped
Spirillum – Spiral
Vibrium – Comma-shaped

(125)Answer : (3)
Hint:
These organisms are also found in marshy habitats.
Solution:
Methanogens are present in the gut of several ruminant animals such as cows and buffaloes. They are responsible for methane production.

(126)Answer : (1)
Hint:
Kingdom Monera includes all prokaryotic organisms.
Solution:
Mode of nutrition in monerans may be chemosynthetic, photosynthetic, saprophytic, symbiotic and parasitic.

(127)Answer : (3)
Solution:
Algae are photosynthetic and fungi are non-photosynthetic.

(128)Answer : (3)
Solution:
Chlamydomonas and *Chlorella* were placed earlier under the kingdom Plantae, due to the presence of cell wall, while *Amoeba* and *Paramoecium* were placed under Animalia, due to the lack of cell wall.

(129)Answer : (1)
Hint:
These are the sole members of kingdom Monera.
Solution:
Bacteria are the most abundant micro-organisms.

(130)Answer : (4)
Solution:
The earliest scientific attempt for classification of organisms was performed by Aristotle.

(131)Answer : (3)
Solution:
Kingdom Monera includes bacteria, some of which are photosynthetic, like cyanobacteria, and they do not have membrane-bound organelles. Kingdom Protista can also have photosynthetic members, but they do have membrane-bound organelles. Therefore, the correct answer is **One** kingdom, which is Monera.

(132)Answer : (2)
Solution:
All single-celled eukaryotes are placed under Protista, but the boundaries of this kingdom are not well defined.

(133)Answer : (3)
Solution:
According to three domain system, archaea and bacteria are the two domains of prokaryotes.

(134)Answer : (3)
Solution:
Majority of bacteria are heterotrophic.

(135)Answer : (1)
Solution:
In the kingdom system of classification proposed by Linnaeus, fungi were placed in the group plantae.

Aakash Medical|IIT-JEE|Foundations

ZOOLOGY

(136)Answer : (1)

Solution:

Except pulmonary arteries, all the arteries carry oxygenated blood. Pulmonary arteries carry deoxygenated blood from the right ventricle to the lungs for oxygenation.

(137)Answer : (1)

Solution:

Opening between right atrium and right ventricle and opening between left atrium and left ventricle is guarded by tricuspid and bicuspid valves, respectively.

(138)Answer : (2)

Solution:

SAN, *i.e*., sino-atrial node is the pacemaker of the human heart situated in the right upper corner of the right atrium of the heart.

(139)Answer : (1)

Solution:

RBCs contain haemoglobin to transport O2.

(140)Answer : (2)

Hint:

Respiratory structures of fishes

Solution:

Special vascularised structures like gills are used by most of the aquatic arthropods. Earthworms use their moist cuticle for respiration. Lungs are respiratory structures of reptiles, birds, and mammals.

(141)Answer : (1)

Solution: Cutaneous respiration.

(142)Answer : (2)

Solution:

On an average, a healthy adult human breathes 12-16 times/minute.

(143)Answer : (3)

Solution:

During hemostasis, prothrombin is converted into thrombin in the presence of prothrombin activator or thrombokinase

(144)Answer : (4)

Solution:

Trachea has C-shaped cartilagenous ring(s) which prevent the collapsing of the trachea.

(145)Answer : (1)

Solution: Under normal conditions 100ml of oxygenated blood delivers 5ml of Oxygen to tissues. Whereas under strenuous conditions oxygen delivery is increased by 3 times making 100ml of oxygenated blood delivers 15ml of Oxygen to tissues. Given that 500ml of blood is being considered 100ml → 15ml 500ml → ? x= (500 X 15)/ 100 = 75ml

(146)Answer : (2)

Hint:

Cardiac output = Stroke volume × Heart rate.

Solution:

SV = EDV – ESV
EDV = 120 mL,
ESV = 50 mL
So, SV = 120 – 50 = 70 mL
SV = Stroke volume
EDV = End diastolic volume
ESV = End systolic volume

(147)Answer : (4)

Solution:

'Dub' is caused by blood turbulence associated with the closure of the semilunar valves at the beginning of ventricular diastole.

(148)Answer : (4)

Solution:

The partial pressure of oxygen in systemic arteries is typically **95 mm Hg.** pO2 in alveoli is 104 mm of Hg. The partial pressure of oxygen in systemic arteries is typically 95 mmHg and in systemic veins it is 40 mmHg.

(149)Answer : (3)

Hint:

Most of the invertebrates have an open type of circulation.

Solution:

All vertebrates and a few invertebrates have a closed circulatory system. Along with vertebrates, the cephalochordates, annelids and cephalopod molluscs have a closed circulatory system. Urochordates, included in the phylum Chordata, also have an open type of blood circulation.

(150)Answer : (4)

Solution:

The clot or coagulation is formed mainly of a network of threads called fibrins in which formed elements of blood are trapped.

(151)Answer : (2)

Solution:

CAD is caused due to deposition of cholesterol, fibrous tissue, *etc*, into coronary artery.

(152)Answer : (1)

Solution:

Chordae tendinae are attached to AV valves and prevent them from collapsing back (prolapse) into the atria during ventricular contractions.

(153)Answer : (1)

Solution:

The movement of air into and out of the lungs occurs due to pressure gradient. There's always a negative pressure inside the lungs compared to the atmospheric pressure.

(154)Answer : (2)

Solution:

Blood is a fluid connective tissue as it does not contains fibroblasts and connective tissue fibres.

(155)Answer : (3)

Solution:

Ribs and sternum return to their original position during normal expiration.

(156)Answer : (2)

Solution:

Volume of air inspired or expired during a normal respiration is called tidal volume. It is approximately 500 mL.

(157)Answer : (1)

Solution:

The flow of the blood is faster in arteries as the blood in them is under great pressure. The walls of the arteries are thick and muscular, as tunica media is thick, having more muscle fibres. Arteries are not collapsible as they have thick walls.

(158)Answer : (4)

Solution:

Tunica externa of blood vessels is an external layer of fibrous connective tissue with collagen fibres. Tunica media is middle layer of smooth muscle and elastic fibres. Tunica intima is an innermost lining of squamous epithellium called endothelium.

(159)Answer : (3)

Solution:

Respiratory Gas	O2	CO_2
Atmospheric Air	159	0.3
Alveoli	104	40
Blood (Deoxygenated)	40	45
Blood (Oxygenated)	95	40
Tissues	40	45

(160)Answer : (4)

Solution:

Shift of the O2-dissociation curve to the right indicates dissociation of oxygen from haemoglobin.

High pCO2 leads to greater H^+concentration and hence, low pH. High pH indicates low pCO2 and hence greater affinity of Hb for O2. This will cause a left shift of the curve.

(161)Answer : (4)

Solution:

Erythropoiesis occurs in red bone marrow in adults.

(162)Answer : (3)

Hint:

Least abundant among all types of WBCs.

Solution:

Basophils secrete histamine, serotonin, heparin, *etc*., and are involved in inflammatory reactions.
Lymphocytes are responsible for immune response of the body. Eosinophils resist infection and are also associated with allergic responses. Neutrophils are phagocytic cells which destroy foreign organisms entering the body.

(163)Answer : (4)

Solution:

A centre present in the pons region of the brain called pneumotaxic centre can moderate the functions of the respiratory rhythm centre.

(164)Answer : (1)

Hint:

Exclude occupational respiratory disorders

Solution:

Asthma is characterised by difficulty in breathing causing wheezing due to inflammation of bronchi and bronchioles.
Emphysema is a chronic disorder in which alveolar walls are damaged due to which respiratory surface is decreased. One of the major causes of this is cigarette smoking.

(165)Answer : (3)

Solution:

In lungs, exchange of gases takes place between alveoli and blood.

(166)Answer : (4)

Solution:

Alveoli are region of actual exchange/diffusion of gases in human lungs. Incomplete rings of hyaline cartilage prevent collapse of primary, secondary, tertiary bronchi and initial bronchioles.

(167)Answer : (1)

Solution:

120/80 mm Hg is the normal systolic/diastolic blood pressure of a healthy adult human male.

(168)Answer : (4)

Hint:

Lymph appears colourless.

Solution:

Lymph consists of plasma and leukocytes. It contains fewer plasma proteins and lower calcium and phosphorous than blood.

(169)Answer : (1)

Solution:

Trachea, which is also known as the windpipe, divides at the level of 5th thoracic vertebra in humans.

(170)Answer : (3)

Solution:

Stroke volume = EDV (end diastolic volume) – ESV (end systolic volume) = 130-60 mL
Stroke volume = 70 mL

Number of beats per minute = $\frac{\text{Cardiac output}}{\text{Stroke volume}}$ = 6300 mL /70 mL

= 90 beats per minute

(171)Answer : (3)

Solution:

Adult human RBCs have an average life span of 120 days.

(172)Answer : (3)

Hint:
Atrio-ventricular septum is formed by connective tissue.
Solution:
A thin, muscular wall called the inter-atrial septum separates the right and left atria, whereas a thick-walled, the inter-ventricular septum, separates the left and the right ventricles.

(173)Answer : (3)
Solution:
Neutrophils and monocytes are both phagocytic cells. Both neutrophils and monocytes are similar in action as they engulf the foreign pathogens.

(174)Answer : (4)
Solution:
Calcium ions play a very important role in clotting.

(175)Answer : (1)
Solution:
Tracheoles are present in cockroach [insects] and help in gaseous exchange. But in humans larynx, bronchi, bronchiole and nasopharynx are present.

(176)Answer : (3)
Solution:
Megakaryocytes are present in bone marrow which are responsible for production of thrombocytes.

(177)Answer : (1)
Hint:
Insect like cockroach possesses tracheal tubes for respiration.
Solution:
Insects have a network of tracheal tubes to transfer atmospheric air into the body.

(178)Answer : (2)
Solution:
Basophils are the least abundant cells among all the WBCs present in the blood of human.

(179)Answer : (3)
Hint:
It leads to relaxation of larger chambers of heart
Solution:
P-wave → Electrical excitation of the atria
QRS-complex → Depolarisation of the ventricles
T-wave → Repolarisation of the ventricles

(180)Answer : (2)
Solution:
Heart, the mesodermally derived organ, is situated in the thoracic cavity in between the two lungs slightly tilted to the left. It has the size of a clenched fist. It is protected by a double walled membranous bag called pericardium, enclosing the pericardial fluid.

Aakash Medical|IIT-JEE|Foundations